Step 3: Mark the points of intersection of the curve $y = 2x^2$ and the line $y = x + 6$. That is, $(-1.5, 4.5)$ and $(2,8)$

Step 4: The x coordinates of the respective points forms the solution set $\{-1.5,2\}$ for $2x^2 - x - 6 = 0$

Example 3.53 Draw the graph of $y = x^2 + 4x + 3$ and hence find the roots of $x^2 + x + 1 = 0$

Solution

Step 1: Draw the graph of $y = x^2 + 4x + 3$ by preparing the table of values as below

x	-4	-3	-2	-1	0	1	2
y	3	0	-1	0	3	8	15

Step 2: To solve $x^2 + x + 1 = 0$, subtract $x^2 + x + 1 = 0$ from $y = x^2 + 4x + 3$

$$\begin{array}{rl} y = & x^2 + 4x + 3 \\ 0 = & x^2 + x + 1 \quad (-) \\ \hline y = & 3x+2 \end{array}$$

The equation represent a straight line. Draw the graph of $y = 3x+2$ forming the table of values as below.

x	-2	-1	0	1	2
y	-4	-1	2	5	8

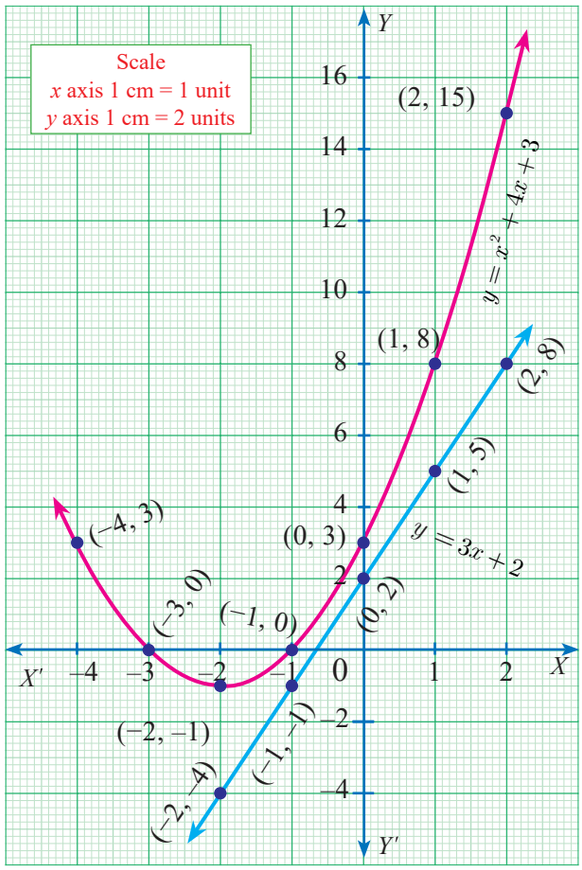


Fig. 3.23

Step 3: Observe that the graph of $y = 3x+2$ does not intersect or touch the graph of the parabola $y = x^2 + 4x + 3$.

Thus $x^2 + x + 1 = 0$ has no real roots.

Example 3.54 Draw the graph of $y = x^2 + x - 2$ and hence solve $x^2 + x - 2 = 0$.

Solution

Step 1: Draw the graph of $y = x^2 + x - 2$ by preparing the table of values as below

x	-3	-2	-1	0	1	2
y	4	0	-2	-2	0	4

Step 2: To solve $x^2 + x - 2 = 0$ subtract $x^2 + x - 2 = 0$ from $y = x^2 + x - 2$

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